PearsonT4 Bootstrap Calibration

Let the observed paired time series be

$$D = \{(t_i, x_i, y_i) : i = 1, \ldots, n\},$$

and let the ordinary Pearson correlation be

$$r = \mathrm{cor}(x, y), \qquad z = \mathrm{atanh}(r).$$

The point estimate is the usual Pearson sample correlation. The bootstrap machinery is used only to construct the confidence interval.

Notation

We use uppercase symbols for bootstrap counts. For T3, the usual nested bootstrap counts are

$$B_1, \quad B_2.$$

For T4, the null-calibration counts are

$$B_1, \quad B_2, \quad B_3,$$

where B_1 is the number of simulated null datasets, B_2 is the number of first-level bootstrap samples within each null dataset, and B_3 is the number of second-level bootstrap samples within each first-level sample.

T3 Bootstrap

PearsonT3 uses a nested moving-block bootstrap. Let B_1 be the number of first-level bootstrap samples and B_2 the number of second-level bootstrap samples.

For $j = 1, \ldots, B_1$, draw a first-level bootstrap sample

$$D_j^*$$

from the observed data and compute

$$r_j^* = \mathrm{cor}(x_j^*, y_j^*), \qquad z_j^* = \mathrm{atanh}(r_j^*).$$

For each first-level sample D_j^*, draw second-level bootstrap samples

$$D_{jk}^{**}, \qquad k = 1, \ldots, B_2,$$

and compute

$$r_{jk}^{**} = \mathrm{cor}(x_{jk}^{**}, y_{jk}^{**}), \qquad z_{jk}^{**} = \mathrm{atanh}(r_{jk}^{**}).$$

The conditional standard error associated with first-level sample j is estimated as

$$s_j = \mathrm{sd}_{k=1}^{B_2}(z_{jk}^{**}).$$

The T3 interval is a Studentized bootstrap interval on Fisher scale, transformed back to correlation scale:

$$\mathrm{CI}_{T3} = \tanh\left(z \pm \lambda_{T3} s_{\mathrm{obs}}\right),$$

where s_{obs} is the estimated standard error for the observed data and λ_{T3} is obtained from the T3 bootstrap calibration.

T4 Null Calibration

PearsonT4 keeps the Pearson point estimate, but replaces the T3 confidence interval construction by a null-calibrated interval. The goal is that a nominal 95 % interval has approximately 5 % two-sided rejection probability when the true correlation is zero.

Let B_1 be the number of null calibration datasets. For $m = 1, \ldots, B_1$, generate an independent-null dataset

$$D_m^0$$

by drawing moving-block bootstrap samples from x and y independently. This preserves the marginal data values and serial dependence approximately, while breaking the contemporaneous association so that the null correlation is

$$\rho = 0.$$

Concretely, one null sample D^0 is constructed from the observed paired series

$$D = \{(x_i, y_i)\}_{i=1}^n$$

as follows. First draw a moving-block bootstrap sample from the observed x series only,

$$x^0 = (x_1^0, \ldots, x_n^0).$$

Then independently draw a moving-block bootstrap sample from the observed y series only,

$$y^0 = (y_1^0, \ldots, y_n^0).$$

Finally pair those two independently resampled series by index:

$$D^0 = \{(x_i^0, y_i^0)\}_{i=1}^n.$$

Thus a null sample D^0 is not generated from an AR model. It is made directly from the observed data by independently block-resampling x and y. The independent block draws impose the null hypothesis $\rho = 0$, while the block structure attempts to preserve each series' own autocorrelation.

For each null dataset D_m^0, run a nested bootstrap with calibration counts B_2 and B_3:

$$D_m^0 \longrightarrow D_{mj}^{0*}, \qquad j = 1, \ldots, B_2,$$

and

$$D_{mj}^{0*} \longrightarrow D_{mjk}^{0**}, \qquad k = 1, \ldots, B_3.$$

For each second-level null bootstrap sample, compute

$$r_{mjk}^{0**} = \mathrm{cor}(x_{mjk}^{0**}, y_{mjk}^{0**}), \qquad z_{mjk}^{0**} = \mathrm{atanh}(r_{mjk}^{0**}).$$

For each first-level null sample, estimate

$$s_{mj}^0 = \mathrm{sd}_{k=1}^{B_3}(z_{mjk}^{0**}).$$

The T4 implementation summarizes these first-level standard errors into a dataset-level null standard error,

$$s_m^0 = \frac{1}{B_2} \sum_{j=1}^{B_2} s_{mj}^0.$$

It then summarizes the null-dataset standard errors into a single T4 scale,

$$s_{T4} = \frac{1}{B_1} \sum_{m=1}^{B_1} s_m^0.$$

Let

$$r_m^0 = \mathrm{cor}(x_m^0, y_m^0), \qquad z_m^0 = \mathrm{atanh}(r_m^0).$$

For candidate lower and upper endpoint values λ_L and λ_U, the null interval for dataset m is

$$\mathrm{CI}_m^0(\lambda_L, \lambda_U) = \left[\tanh\left(z_m^0 + t(\lambda_L) s_{T4}\right), \tanh\left(z_m^0 - t(\lambda_U) s_{T4}\right)\right],$$

where $t(\lambda)$ is the Student-t quantile used by PearsonT. The two null tail error rates are estimated separately:

$$\widehat{\alpha}_L(\lambda_L) = \frac{1}{B_1}\sum_{m=1}^{B_1} I\{L_m^0(\lambda_L) > 0\},$$

and

$$\widehat{\alpha}_U(\lambda_U) = \frac{1}{B_1}\sum_{m=1}^{B_1} I\{U_m^0(\lambda_U) < 0\}.$$

For a 95 % two-sided confidence interval, PearsonT4 uses the one-tail level

$$\alpha = 0.025.$$

It chooses λ_L and λ_U so that

$$\widehat{\alpha}_L(\lambda_L) \approx \alpha, \qquad \widehat{\alpha}_U(\lambda_U) \approx \alpha.$$

The final T4 interval for the observed data is

$$\mathrm{CI}_{T4} = \left[\tanh\left(z + t(\lambda_L)s_{T4}\right), \tanh\left(z - t(\lambda_U)s_{T4}\right)\right].$$

Thus the reported T4 interval uses the same null-path scale s_{T4} that was used to calibrate the null rejection probability. It does not use the observed-data T3 bootstrap standard error. Because λ_L and λ_U are chosen separately, the T4 interval need not be symmetric on Fisher scale.

Computational Cost

The null-calibration part has cost approximately

$$B_1 B_2 B_3.$$

For example, with

$$B_1 = 1000, \qquad B_2 = 100, \qquad B_3 = 50,$$

the null calibration costs roughly

$$1000 \cdot 100 \cdot 50 = 5 \times 10^6$$

correlation calculations.